ADV: Functions (Adv), F1 Working with Functions (Adv)

Quadratics and Cubic Functions (Y11)
Further Functions and Relations (Y11)

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Exam Equivalent Time: 133.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

Questions

1. Functions, 2ADV F1 2013 HSC 1 MC

What are the solutions of $2x^2 - 5x - 1 = 0$?

(A) $x = \dfrac{-5 \pm \sqrt{17}}{4}$

(B) $x = \dfrac{5 \pm \sqrt{17}}{4}$

(C) $x = \dfrac{-5 \pm \sqrt{33}}{4}$

(D) $x = \dfrac{5 \pm \sqrt{33}}{4}$

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2. Functions, 2ADV F1 2014 HSC 6 MC

Which expression is a factorisation of $8x^3 + 27$?

(A) $(2x - 3)(4x^2 + 12x - 9)$

(B) $(2x + 3)(4x^2 - 12x + 9)$

(C) $(2x - 3)(4x^2 + 6x - 9)$

(D) $(2x + 3)(4x^2 - 6x + 9)$

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3. Functions, 2ADV F1 2017 HSC 2 MC

Which expression is equal to $3x^2 - x - 2$?

(A) $(3x - 1)(x + 2)$

(B) $(3x + 1)(x - 2)$

(C) $(3x - 2)(x + 1)$

(D) $(3x + 2)(x - 1)$

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4. Functions, 2ADV F1 2019 HSC 2 MC

What values of x satisfy $4 - 3x \leq 12$?

(A) $x \leq -\dfrac{16}{3}$

(B) $x \geq -\dfrac{16}{3}$

(C) $x \leq -\dfrac{8}{3}$

(D) $x \geq -\dfrac{8}{3}$

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5. Functions, 2ADV F1 2013 HSC 3 MC

Which inequality defines the domain of the function $f(x) = \dfrac{1}{\sqrt{x+3}}$?

(A) $x > -3$

(B) $x \geq -3$

(C) $x < -3$

(D) $x \leq -3$

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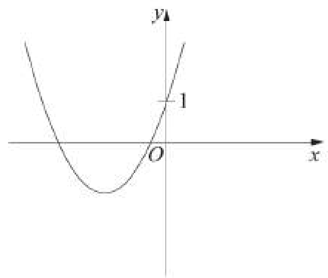
6. Functions, 2ADV F1 2020 HSC 5 MC

Which of the following could represent the graph of $y = -x^2 + bx + 1$, where $b > 0$?

A.



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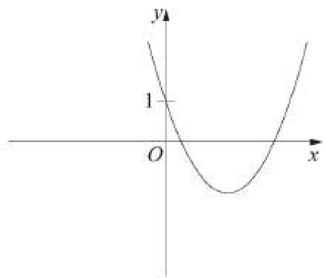

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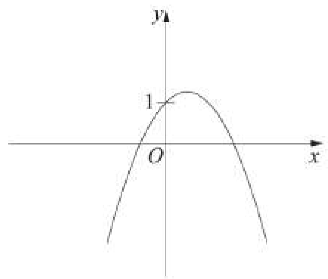
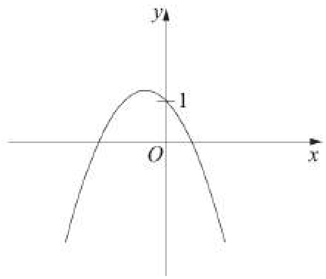


D.



7. Functions, 2ADV F1 2020 HSC 1 MC

Which inequality gives the domain of $y = \sqrt{2x - 3}$?

A. $x < \dfrac{3}{2}$

B. $x > \dfrac{3}{2}$

C. $x \leq \dfrac{3}{2}$

D. $x \geq \dfrac{3}{2}$

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8. Functions, 2ADV F1 SM-Bank 21 MC

A circle with centre $(a, \; -2)$ and radius 5 units has equation

$x^2 - 6x + y^2 + 4y = b$ where a and b are real constants.

The values of a and b are respectively

A. −3 and 38

B. 3 and 12

C. −3 and −8

D. 3 and 18

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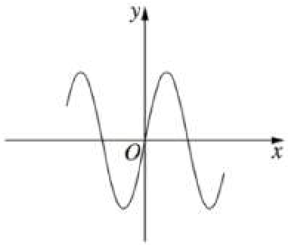
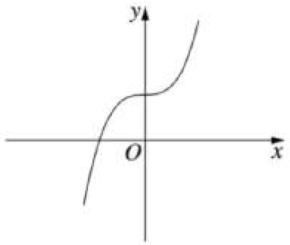
9. Functions, 2ADV F1 2016 HSC 4 MC

Which diagram shows the graph of an odd function?

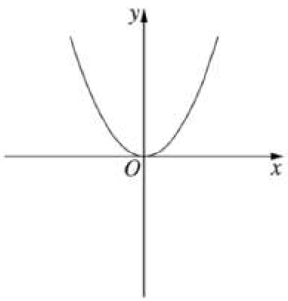
(A)



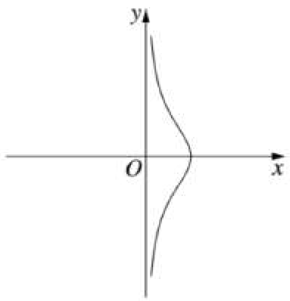
(B)



(C)



(D)



10. Functions, 2ADV F1 2006 HSC 1b

Factorise $2x^2 + 5x - 3$. *(2 marks)*

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11. Functions, 2ADV F1 2007 HSC 1b

Solve $2x - 5 > -3$ and graph the solution on a number line. *(2 marks)*

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12. Functions, 2ADV F1 2007 HSC 1e

Factorise $2x^2 + 5x - 12$. *(2 marks)*

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13. Functions, 2ADV F1 2009 HSC 1c

Solve $|\,x + 1\,| = 5$. *(2 marks)*

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14. Functions, 2ADV F1 2010 HSC 1d

Solve $|\,2x + 3\,| = 9$. *(2 marks)*

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15. Functions, 2ADV F1 2016 HSC 11e

Find the points of intersection of $y = -5 - 4x$ and $y = 3 - 2x - x^2$. *(3 marks)*

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16. Functions, 2ADV F1 2006 HSC 1e

Solve $3 - 5x \leq 2$. *(2 marks)*

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17. Functions, 2ADV F1 2010 HSC 1a

Solve $x^2 = 4x$. *(2 marks)*

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18. Functions, 2ADV F1 2011 HSC 1b

Simplify $\dfrac{n^2 - 25}{n - 5}$. *(1 mark)*

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19. Functions, 2ADV F1 2011 HSC 1e

Solve $2 - 3x \leq 8$. *(2 marks)*

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20. Functions, 2ADV F1 2012 HSC 11a

Factorise $2x^2 - 7x + 3$ *(2 marks)*

21. Functions, 2ADV F1 2014 HSC 11b

Factorise $3x^2 + x - 2$. *(2 marks)*

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22. Functions, 2ADV F1 2015 HSC 11b

Factorise fully $3x^2 - 27$. *(2 marks)*

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23. Functions, 2ADV F1 2017 HSC 11g

Solve $|3x - 1| = 2$. *(2 marks)*

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24. Functions, 2ADV F1 2018 HSC 11b

Solve $1 - 3x > 10$. *(2 marks)*

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25. Functions, 2ADV F1 2008 HSC 1d

Solve $|4x - 3| = 7$. *(2 marks)*

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26. Functions, 2ADV F1 SM-Bank 33

i. State the domain and range of $y = -\sqrt{12 - x^2}$. *(2 marks)*

ii. Sketch the graph. *(1 mark)*

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27. Functions, 2ADV F1 SM-Bank 41

Find the values of x for which $|x + 1| = 5$. *(2 marks)*

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28. Functions, 2ADV F1 SM-Bank 42

Find the values of x for which $|x - 3| = 1$. *(2 marks)*

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29. Functions, 2ADV F1 SM-Bank 44

Solve $|x - 2| = 3$. *(2 marks)*

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30. Functions, 2ADV F1 SM-Bank 37

Find all values of x for which $|x - 4| = \dfrac{x}{2} + 7$. *(3 marks)*

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31. Functions, 2ADV F1 2016 HSC 11a

Sketch the graph of $(x - 3)^2 + (y + 2)^2 = 4$. *(2 marks)*

32. Functions, 2ADV F1 SM-Bank 35

i. Sketch the function $y = f(x)$ where $f(x) = (x - 1)^3$ on a number plane, labelling all intercepts. *(1 mark)*

ii. On the same graph, sketch $y = -f(-x)$. Label all intercepts. *(2 marks)*

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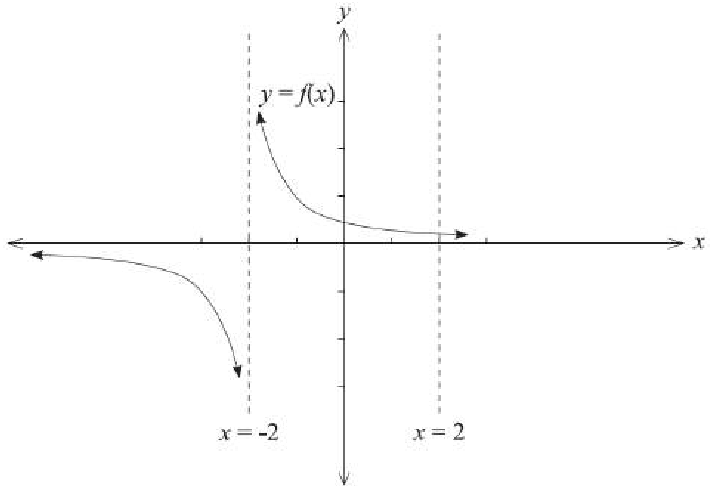
33. Functions, 2ADV F1 SM-Bank 36

Consider the function $f(x) = \dfrac{1}{x + 2}$.



i. Sketch the graph $y = f(-x)$. *(2 marks)*

ii. On the same graph, sketch $y = -f(x)$. *(2 marks)*

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34. Functions, 2ADV F1 2019 HSC 13e

i. Sketch the graph of $y = |x - 1|$ for $-4 \le x \le 4$. *(1 mark)*

ii. Using the sketch from part i, or otherwise, solve $|x - 1| = 2x + 4$. *(2 marks)*

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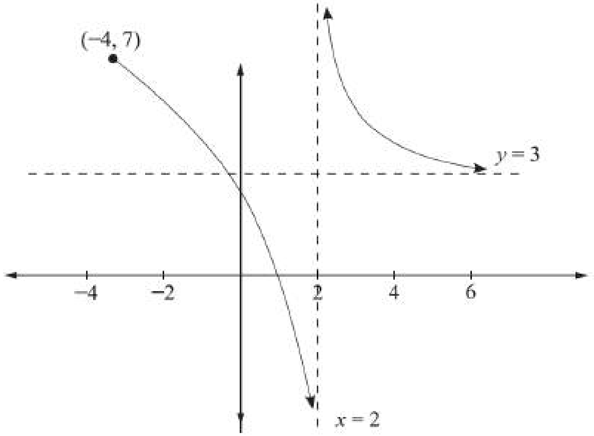
35. Functions, 2ADV F1 EQ-Bank 6

The graph of $f(x)$ is shown below. It has asymptotes at $y=3$ and $x=2$.



Using interval notation, state the domain and range of $f(x)$. *(2 marks)*

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36. Functions, 2ADV F1 EQ-Bank 7

The current of an electrical circuit, measured in amps (A), varies inversely with its resistance, measured in ohms (R).

When the resistance of a circuit is 28 ohms, the current is 3 amps.

What is the current when the resistance is 8 ohms? *(2 marks)*

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37. Functions, 2ADV F1 EQ-Bank 8

Jacques is a marine biologist and finds that the mass of a crab is directly proportional to the cube of the diameter of its shell.

If a crab with a shell diameter of 15 cm weighs 680 grams, what will be the diameter of a crab that weighs 1.1 kilograms? Give your answer to 1 decimal place. *(2 marks)*

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38. Functions, 2ADV F1 SM-Bank 23

Find the values of k for which the expression $x^2-3x+(4-2k)$ is always positive. *(3 marks)*

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39. Functions, 2ADV F1 EQ-Bank 26

Fuifui finds that for Giant moray eels, the mass of an eel is directly proportional to the cube of its length.

An eel of this species has a length of 25 cm and a mass of 4350 grams.

What is the expected length of a Giant moray eel with a mass of 6.2 kg? Give your answer to one decimal place. *(3 marks)*

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40. Functions, 2ADV F1 EQ-Bank 27

The stopping distance of a car on a certain road, once the brakes are applied, is directly proportional to the square of the speed of the car when the brakes are first applied.

A car travelling at 70 km/h takes 58.8 metres to stop.

How far does it take to stop if it is travelling at 105 km/h? *(3 marks)*

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41. Functions, 2ADV F1 SM-Bank 32

Find the centre and radius of the circle with the equation

$x^2-12x+y^2+2y-12=0$ *(2 marks)*

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42. Functions, 2ADV F1 2010 HSC 1c

Write down the equation of the circle with centre $(-1, 2)$ and radius 5. *(1 mark)*

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43. Functions, 2ADV F1 2010 HSC 1g

Let $f(x)=\sqrt{x-8}$. What is the domain of $f(x)$? *(1 mark)*

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44. Functions, 2ADV F1 2017 HSC 11h

Find the domain of the function $f(x)=\sqrt{3-x}$. *(2 marks)*

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45. Functions, 2ADV F1 2020 HSC 24

The circle of $x^2-6x+y^2+4y-3=0$ is reflected in the x-axis.

Sketch the reflected circle, showing the coordinates of the centre and the radius. *(3 marks)*

Worked Solutions

1. Functions, 2ADV F1 2013 HSC 1 MC

$2x^2 - 5x - 1 = 0$

$\text{Using } x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4 \times 2 \times (-1)}}{2 \times 2}$$

$$= \frac{5 \pm \sqrt{25 + 8}}{4}$$

$$= \frac{5 \pm \sqrt{33}}{4}$$

$\Rightarrow D$

2. Functions, 2ADV F1 2014 HSC 6 MC

$8x^3 + 27$

$= (2x)^3 + 3^3$

$= (2x + 3)(4x^2 - 6x + 9)$

COMMENT: Factorising a cubic is only examinable with scaffolding, as provided here by expanding the answer options.

$\Rightarrow D$

3. Functions, 2ADV F1 2017 HSC 2 MC

$3x^2 - x - 2$

$= (3x + 2)(x - 1)$

$\Rightarrow D$

4. Functions, 2ADV F1 2019 HSC 2 MC

$4 - 3x \le 12$

$-3x \le 8$

$x \ge -\dfrac{8}{3}$

$\Rightarrow D$

Worked Solutions

5. Functions, 2ADV F1 2013 HSC 3 MC

$\text{Given } f(x) = \dfrac{1}{\sqrt{x+3}}$

$\text{We know } (x+3) > 0$

$x > -3$

$\therefore$ The domain of $f(x)$ is $f(x) > -3$

$\Rightarrow A$

6. Functions, 2ADV F1 2020 HSC 5 MC

$y = -x^2 + bx + 1$

$\dfrac{dy}{dx} = -2x + b$

S.P. occurs when $\dfrac{dy}{dx} = 0$:

$-2x + b = 0$

$x = \dfrac{b}{2}$

Since $b > 0$, SP occurs when $x > 0$

$\Rightarrow C$

7. Functions, 2ADV F1 2020 HSC 1 MC

Domain exists when:

$2x - 3 \ge 0$

$2x \ge 3$

$x \ge \dfrac{3}{2}$

$\Rightarrow D$

8. Functions, 2ADV F1 SM-Bank 21 MC

$x^2 - 6x + y^2 + 4y = b$

Completing the squares:

$$x^2 - 6x + 3^2 - 9 + y^2 + 4y + 2^2 - 4 = b$$
$$(x-3)^2 + (y+2)^2 - 13 = b$$
$$(x-3)^2 + (y+2)^2 = b + 13$$

$\therefore a = 3$

$\therefore b + 13 = 25 \quad \Rightarrow \quad b = 12$

$\Rightarrow B$

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9. Functions, 2ADV F1 2016 HSC 4 MC

Odd functions occur when:

$f(x) = -f(x)$

Graphically, this occurs when a function has symmetry when rotated 180° about the origin.

$\Rightarrow A$

♦ Mean mark 38%.

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10. Functions, 2ADV F1 2006 HSC 1b

$$2x^2 + 5x - 3$$
$$= (2x-1)(x+3)$$

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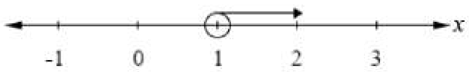
11. Functions, 2ADV F1 2007 HSC 1b

$$2x - 5 > -3$$
$$2x > 2$$
$$x > 1$$



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12. Functions, 2ADV F1 2007 HSC 1e

$$2x^2 + 5x - 12$$
$$= (2x-3)(x+4)$$

---

13. Functions, 2ADV F1 2009 HSC 1c

$|x+1| = 5$

$$(x+1) = 5 \qquad -(x+1) = 5$$
$$x = 4 \qquad -x - 1 = 5$$
$$x = -6$$

$\therefore x = 4 \text{ or } -6$

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14. Functions, 2ADV F1 2010 HSC 1d

$|2x+3| = 9$

$$2x + 3 = 9 \qquad -(2x+3) = 9$$
$$2x = 6 \qquad -2x - 3 = 9$$
$$x = 3 \qquad -2x = 12$$
$$x = -6$$

$\therefore x = 3 \text{ or } -6$

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15. Functions, 2ADV F1 2016 HSC 11e

$y = 3 - 2x - x^2$

Substitute $y = -5 - 4x$ into equation

$$-5 - 4x = 3 - 2x - x^2$$
$$x^2 - 2x - 8 = 0$$
$$(x-4)(x+2) = 0$$

$\therefore x = 4 \text{ or } -2$

When $x = 4$, $y = -5 - 4(4) = -21$

When $x = -2$, $y = -5 - 4(-2) = 3$

$\therefore$ Intersection at $(4, -21)$ and $(-2, 3)$

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16. Functions, 2ADV F1 2006 HSC 1e

$$\begin{aligned} 3-5x &\leq 2 \\ -5x &\leq -1 \\ x &\geq \frac{1}{5} \end{aligned}$$

17. Functions, 2ADV F1 2010 HSC 1a

$$\begin{aligned} x^2 &= 4x \\ x^2-4x &= 0 \\ x(x-4) &= 0 \end{aligned}$$

$$\therefore\ x = 0 \text{ or } 4$$

18. Functions, 2ADV F1 2011 HSC 1b

$$\begin{aligned} \frac{n^2-25}{n-5} &= \frac{(n-5)(n+5)}{n-5} \\ &= n+5 \end{aligned}$$

19. Functions, 2ADV F1 2011 HSC 1e

$$\begin{aligned} 2-3x &\leq 8 \\ -3x &\leq 6 \\ x &\geq -\frac{6}{3} \\ x &\geq -2 \end{aligned}$$

20. Functions, 2ADV F1 2012 HSC 11a

$$\begin{aligned} &2x^2-7x+3 \\ &= (2x-1)(x-3) \end{aligned}$$

STRATEGY: Check your answer by expanding factors.

21. Functions, 2ADV F1 2014 HSC 11b

$$\begin{aligned} &3x^2+x-2 \\ &= (3x-2)(x+1) \end{aligned}$$

22. Functions, 2ADV F1 2015 HSC 11b

$$\begin{aligned} 3x^2-27 &= 3(x^2-9) \\ &= 3(x+3)(x-3) \end{aligned}$$

23. Functions, 2ADV F1 2017 HSC 11g

$$|3x-1| = 2$$

$$\begin{aligned} 3x-1 &= 2 \quad \text{or} \quad & -(3x-1) &= 2 \\ x &= 1 & -3x &= 1 \\ & & x &= -\frac{1}{3} \end{aligned}$$

$$\therefore\ x = 1 \text{ or } -\frac{1}{3}$$

MARKER'S COMMENT: Note that both conditions must be satisfied! Dealing with negative signs and division for inequalities produced many errors.

24. Functions, 2ADV F1 2018 HSC 11b

$$\begin{aligned} 1-3x &> 10 \\ -3x &> 9 \\ x &< -3 \end{aligned}$$

25. Functions, 2ADV F1 2008 HSC 1d

$$|\,4x-3\,| = 7$$

$$\begin{aligned} 4x-3 &= 7 & -(4x-3) &= 7 \\ 4x &= 10 & -4x+3 &= 7 \\ x &= \frac{5}{2} & -4x &= 4 \\ & & x &= -1 \end{aligned}$$

$$\therefore\ x = \frac{5}{2} \text{ or } -1$$

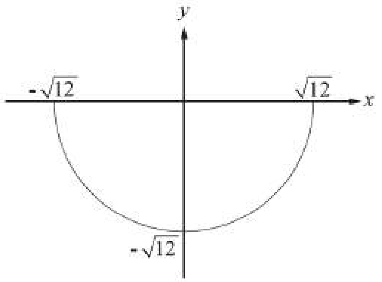
26. Functions, 2ADV F1 SM-Bank 33

i. $y = -\sqrt{12 - x^2}$

$$\text{Domain: } -\sqrt{12} \le x \le \sqrt{12}$$

$$\text{Range: } -\sqrt{12} \le y \le 0$$

ii.



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27. Functions, 2ADV F1 SM-Bank 41

$$|\,x + 1\,| = 5$$

$$(x+1)^2 = 5^2$$

$$x^2 + 2x + 1 = 25$$

$$x^2 + 2x - 24 = 0$$

$$(x+6)(x-4) = 0$$

$$\therefore x = 4 \text{ or } -6$$

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28. Functions, 2ADV F1 SM-Bank 42

$$|\,x - 3\,| = 1$$

Method 1

$$(x-3)^2 = 1$$

$$x^2 - 6x + 9 = 1$$

$$x^2 - 6x + 8 = 0$$

$$(x-4)(x-2) = 0$$

$$\therefore x = 2 \text{ or } 4$$

Method 2

$$\begin{aligned} (x-3) &= 1 & -(x-3) &= 1 \\ x &= 4 & -x + 3 &= 1 \\ & & x &= -2 \end{aligned}$$

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29. Functions, 2ADV F1 SM-Bank 44

$$|\,x - 2\,| = 3$$

$$(x-2)^2 = 3^2$$

$$\left(x^2 - 4x + 4\right) = 9$$

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

$$\therefore x = -1 \text{ or } 5$$

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30. Functions, 2ADV F1 SM-Bank 37

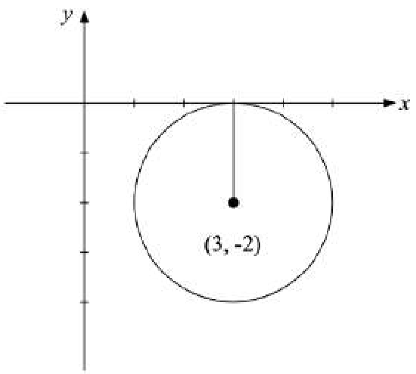
$$\begin{aligned} x - 4 &= \frac{x}{2} + 7 \quad \text{or} & -(x-4) &= \frac{x}{2} + 7 \\ 2x - 8 &= x + 14 & -2x + 8 &= x + 14 \\ x &= 22 & 3x &= -6 \\ & & x &= -2 \end{aligned}$$

$$\therefore x = 22 \text{ or } -2$$

31. Functions, 2ADV F1 2016 HSC 11a

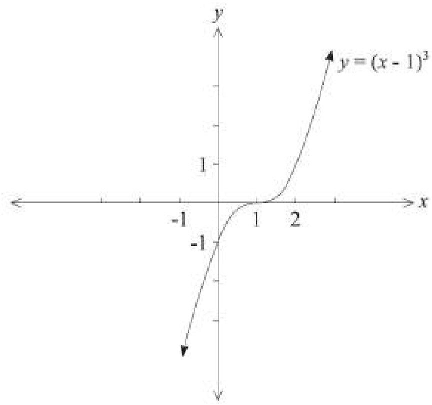
$(x-3)^2+(y+2)^2=4$ is a circle,

centre $(3,\ -2)$, radius 2.



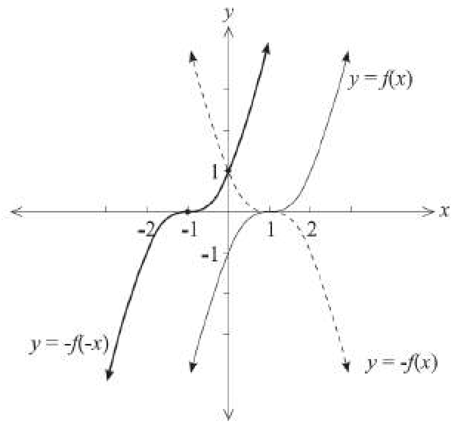
32. Functions, 2ADV F1 SM-Bank 35

i. $y=(x-1)^3 \Rightarrow y=x^3$ shifted 1 unit to the right.



ii. $y=-f(x) \Rightarrow$ reflect $y=(x-1)^3$ in x-axis.

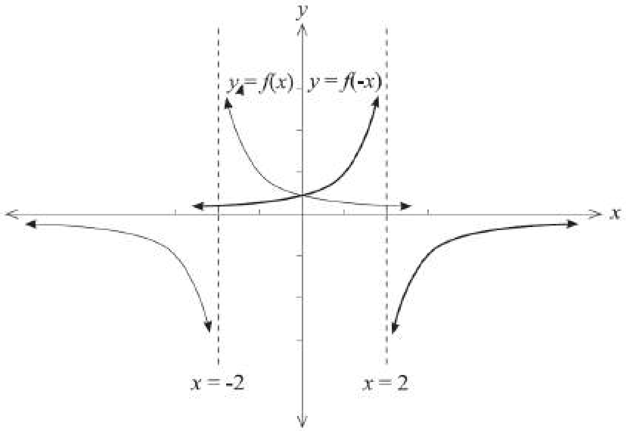
$y=-f(-x) \Rightarrow$ reflect $y=-f(x)$ in y-axis.



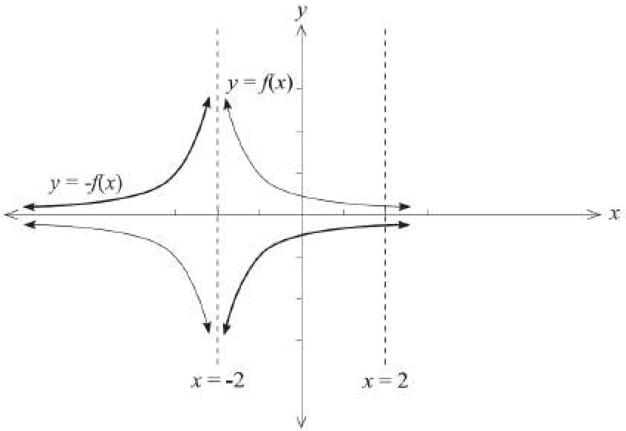
33. Functions, 2ADV F1 SM-Bank 36

i. Sketch $y = \dfrac{1}{x+2}$

$y = f(-x) \Rightarrow$ reflect $y = \dfrac{1}{x+2}$ in the y-axis.

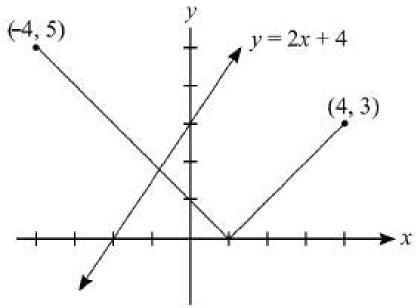


ii. $y = -f(x) \Rightarrow$ reflect $y = \dfrac{1}{x+2}$ in the x-axis.



34. Functions, 2ADV F1 2019 HSC 13e

(i)



ii. By inspection, intersection when $x = -1$

Test:

$$|-1-1| = -2+4$$
$$2 = 2$$

$\therefore$ Intersection at $(-1, 2)$

35. Functions, 2ADV F1 EQ-Bank 6

Domain: $[-4, 2) \cup (2, \infty)$

Range: $(-\infty, \infty)$

36. Functions, 2ADV F1 EQ-Bank 7

$$A \propto \frac{1}{R}$$
$$A = \frac{k}{R}$$

When $A = 3,\ R = 28$

$$3 = \frac{k}{28}$$
$$k = 84$$

Find A *when* $R = 8$:

$$A = \frac{84}{8}$$
$$= 10.5$$

37. Functions, 2ADV F1 EQ-Bank 8

$$M \propto d^3$$
$$M = kd^3$$

When $M = 680,\ d = 15$

$680 = k \times 15^3$
$k = 0.201481...$

Find d when $M = 1100$:

$$1100 = 0.20148... \times d^3$$
$$d = \sqrt[3]{\frac{1100}{0.20148...}}$$
$$= 17.608...$$
$$= 17.6 \text{ cm} \ \ (\text{to 1 d.p.})$$

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38. Functions, 2ADV F1 SM-Bank 23

$x^2 - 3x + (4 - 2k) > 0$

$x^2 - 3x + (4 - 2k) = 0$ is a concave up parabola

$\Rightarrow$ Always positive (no roots) if $\Delta < 0$

$$b^2 - 4ac < 0$$
$$(-3)^2 - 4 \cdot 1 \cdot (4 - 2k) < 0$$
$$9 - 16 + 8k < 0$$
$$8k < 7$$
$$k < \frac{7}{8}$$

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39. Functions, 2ADV F1 EQ-Bank 26

Mass $\propto$ length^3

$m = kl^3$

Find k:

$$4350 = k \times 25^3$$
$$k = \frac{4350}{25^3}$$
$$= 0.2784$$

Find l when $m = 6200$:

$$6200 = 0.2784 \times l^3$$
$$l^3 = \frac{6200}{0.2784}$$
$$\therefore l = 28.13\ldots$$
$$= 28.1 \text{ cm} \ \ (\text{to 1 d.p.})$$

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40. Functions, 2ADV F1 EQ-Bank 27

Let $d =$ stopping distance

$$d \propto s^2$$
$$d = ks^2$$

Find k,

$$58.8 = k \times 70^2$$
$$k = \frac{58.8}{70^2}$$
$$= 0.012$$

Find d when $s = 105$:

$$d = 0.012 \times 105^2$$
$$= 132.3 \text{ metres}$$

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41. Functions, 2ADV F1 SM-Bank 32

$$x^2 - 12x + y^2 + 2y - 12 = 0$$
$$(x-6)^2 + (y+1)^2 - 36 - 1 - 12 = 0$$
$$(x-6)^2 + (y+1)^2 = 49$$

$\therefore$ Centre $(6, -1)$

$\therefore$ Radius $= 7$

42. Functions, 2ADV F1 2010 HSC 1c

Circle with centre $(-1, 2)$, $r = 5$

$$(x+1)^2 + (y-2)^2 = 25$$

MARKER'S COMMENT: Expanding this equation is not necessary!

43. Functions, 2ADV F1 2010 HSC 1g

$f(x) = \sqrt{x - 8}$

Domain where

$(x-8) \geq 0$

$x \geq 8$

♦ *Mean mark 49%.*
MARKER'S COMMENT: $x > 8$ was a common incorrect answer.

44. Functions, 2ADV F1 2017 HSC 11h

Solution 1

Domain of $f(x) = \sqrt{3 - x}$

$3 - x \geq 0$

$x \leq 3$

Note domain can also be expressed as: $(-\infty, 3]$

45. Functions, 2ADV F1 2020 HSC 24

$$x^2 - 6x + y^2 + 4y - 3 = 0$$
$$x^2 - 6x + 9 + y^2 + 4y + 4 - 16 = 0$$
$$(x-3)^2 + (y+2)^2 = 16$$

$\Rightarrow$ Original circle has centre $(3, -2)$, radius $= 4$

Reflect in x-axis:

Centre $(3, -2) \rightarrow (3, 2)$

♦ *Mean mark 48%.*